M. Chinni Krishna Reddy

192411192

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery.**(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: **Total Marks: 20**

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: **DOM Structure Analysis**

Consider the following DOM tree representation:

```
<html>
  <head>
  <title>TestPage</title>
</head>
  <body>
  <div>
  <p>HelloWorld
  </div>
 </body>
 </html>
```

Questions:

1. Identify missing or improperly closed elements.

Answer: The element is not closed (missing before). The text 'HelloWorld' is also missing a space, and the document has no <!DOCTYPE html> or lang attribute. (is optional in HTML5, but closing it is good practice.)

2. Analyze how the browser will interpret this structure.

Answer: HTML parsers are error-tolerant. When the browser reaches while is still open, it implicitly closes the . The resulting DOM is html > head > title, and body > div > p > 'HelloWorld'. Without a DOCTYPE the browser switches to quirks mode.

3. Determine the impact on rendering and layout.

Answer: Here the text still renders because the paragraph is auto-closed. However, quirks mode can cause inconsistent box model, fonts and spacing; in bigger pages an unclosed causes unexpected nesting, wrong margins, broken CSS selectors/JavaScript traversal, and the page fails validation.

4. Suggest corrections to make the DOM valid.

Answer: Add the DOCTYPE, close the tag and fix the text:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>TestPage</title>
</head>
<body>
  <div>
    <p>Hello World</p>
  </div>
```

```
</body>
</html>
```

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.

Answer: GET appends data to the URL as a query string (?user=admin&pass=1234), so it is visible in the address bar, browser history and server/proxy logs, and has length limits. POST sends data in the HTTP request body, which is not shown in the URL and has no practical size limit.

2. Analyze which method is more suitable for login and why.

Answer: POST is more suitable. Credentials stay out of the URL, so they are not saved in history, bookmarks, logs or Referer headers; POST responses are not cached and it suits state-changing actions like creating a session. POST must still be used over HTTPS, because the body is plain text otherwise.

3. Identify security risks in using GET for authentication.

Answer: Password exposed in the URL (shoulder-surfing); stored in browser history and bookmarks; recorded in server, proxy and ISP logs; leaked to other sites via the Referer header; cached by browsers/proxies; easily shared or replayed by copying the link; and easier to abuse with CSRF-style attacks.

4. Suggest best practices for secure login implementation.

Answer: Use POST over HTTPS/TLS only (with HSTS); store passwords as salted hashes (bcrypt/Argon2/PBKDF2); use CSRF tokens; use secure session cookies (HttpOnly, Secure, SameSite) and regenerate the session ID after login; validate/sanitise input to prevent SQL injection and XSS; add rate limiting, lockout or CAPTCHA against brute force; use multi-factor authentication; show generic error messages; never put credentials in URLs or logs.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

Questions:

1. Identify missing closing tags.

Answer: The <h1> tag is missing </h1> and the tag is missing . (<img> is a void element and needs no closing tag in HTML5.)

2. Analyze issues with the `<img>` tag.

Answer: The <img> has no alt attribute, which is required for valid HTML and accessibility (screen readers and fallback text). It also has no width/height, which can cause layout shift, and the relative path 'image.jpg' shows a broken icon if the file is not found. In XHTML it must also be self-closed ().

3. Determine how browsers will handle these errors.

Answer: Browsers use HTML5 error recovery: <h1> is implicitly closed when starts, and is closed at </body>, so the page still renders (heading, paragraph, image) without any visible error. A broken image shows an icon with no alt text. A strict XHTML parser would stop with a parse error.

4. Suggest a corrected version of the code.

Answer: Corrected code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Sample</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a paragraph</p>
  <img src="image.jpg" alt="Sample image"
width="300" height="200">
</body>
</html>
```

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser

5. Response is returned
6. Browser renders page

2. DNS lookup occurs

3. HTTP request is sent

4. Server processes request

Questions:

1. Interpret each step in the communication flow.

Answer: 1) The browser parses the URL (protocol, domain, path) and checks its cache. 2) DNS translates the domain name into an IP address. 3) A TCP connection (plus TLS handshake for HTTPS) is made and the HTTP request (e.g., GET /index.html) is sent. 4) The server runs application logic, reads files/database and prepares the content. 5) The server returns an HTTP response with status code, headers and body. 6) The browser builds the DOM/CSSOM, loads CSS/JS/images, runs scripts and paints the page.

2. Identify where delays can occur.

Answer: DNS lookup (slow or uncached resolver); TCP/TLS handshake and network latency; bandwidth/congestion; slow server processing (heavy code or database queries); large responses or many extra requests; and browser rendering (render-blocking CSS/JS, heavy DOM).

3. Analyze the role of DNS in the process.

Answer: DNS is the internet's phonebook: it maps a domain name (www.example.com) to the server's IP address, which browsers need to connect. If DNS fails, the server cannot be found and the request fails. It is the first network step, so its speed affects page load time; results are cached (browser, OS, ISP) according to TTL to avoid repeated lookups.

4. Suggest optimizations to improve performance.

Answer: Use DNS caching, a fast DNS provider and dns-prefetch/preconnect; use a CDN; enable HTTP/2 or HTTP/3 and keep-alive; enable GZIP/Brotli compression; minify CSS/JS/HTML; compress and lazy-load images; use browser caching (Cache-Control, ETag); reduce requests and render-blocking resources (async/defer); optimise server code, database queries and server-side caching.

Code of Conduct:

I _M.Chinni Krishna reddy(192411192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

m.chinni Krishna reddy

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks Obtaine d
Understandin g of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	20
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	20
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	17

RUBRICS FOR CALCULATION:

Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	17
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	10
total						80/100